

26/08/2026

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EXAM: WRITTE EXAM

- OPEN QUESTIONS

- EXERCISES.

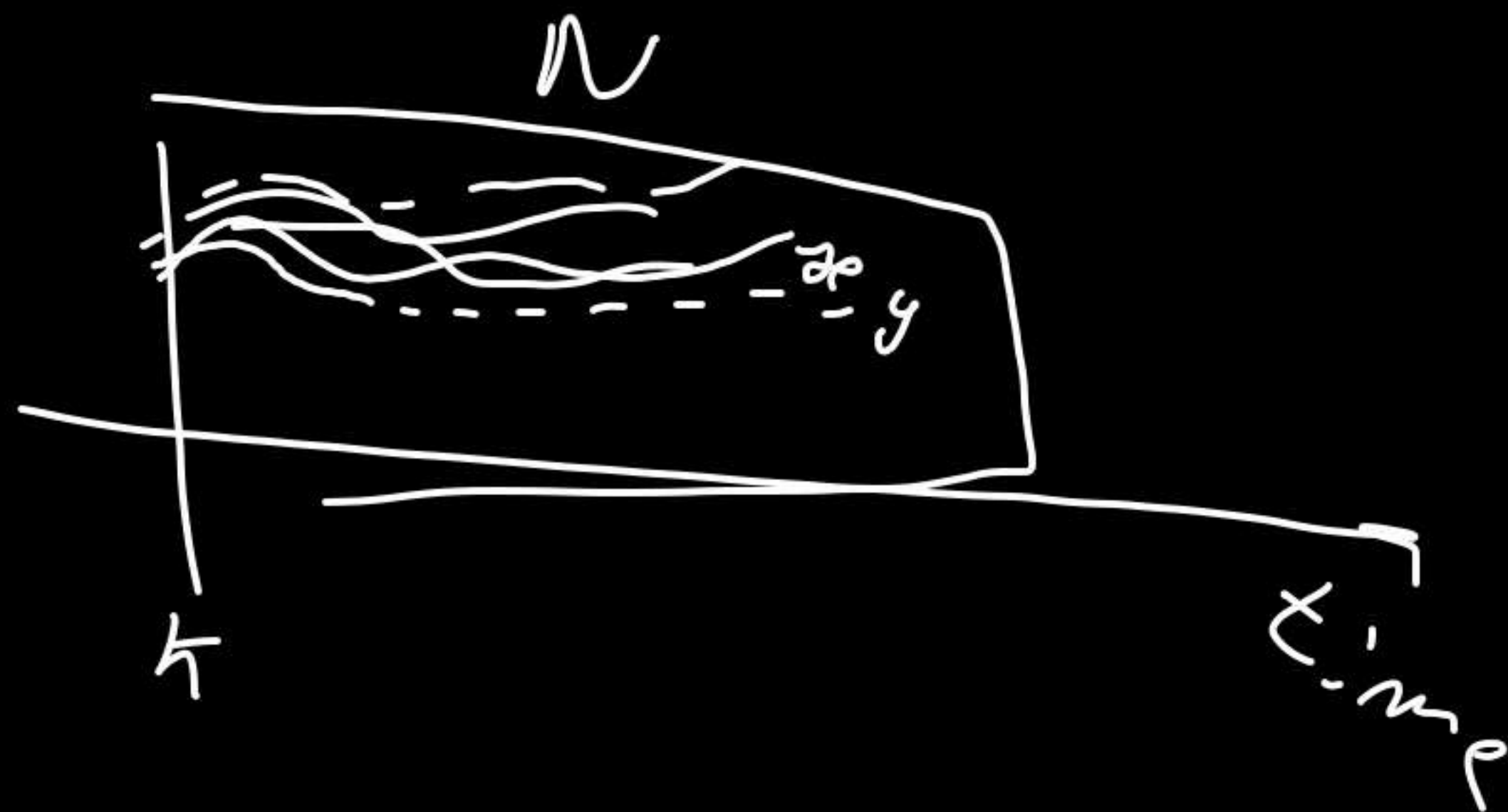
NOMINAL MPC

- PERFECT MODEL

→ SIMPLER ANALYSIS OF THE
THREE PROPERTIES

- PERFECT PREDICTIONS

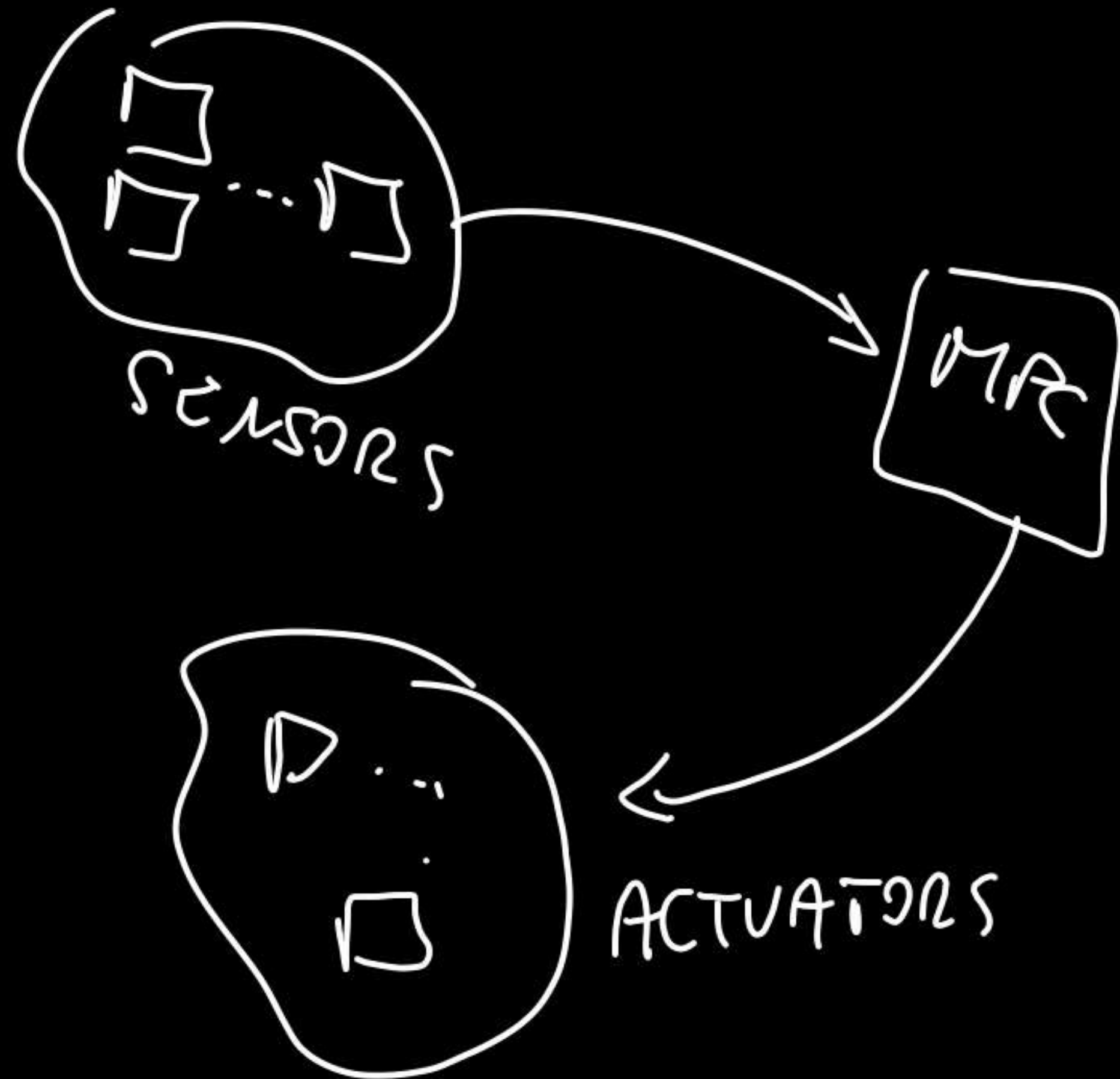
ROBUST MPC (TUBE MPC)



PROBABILISTIC MPC.

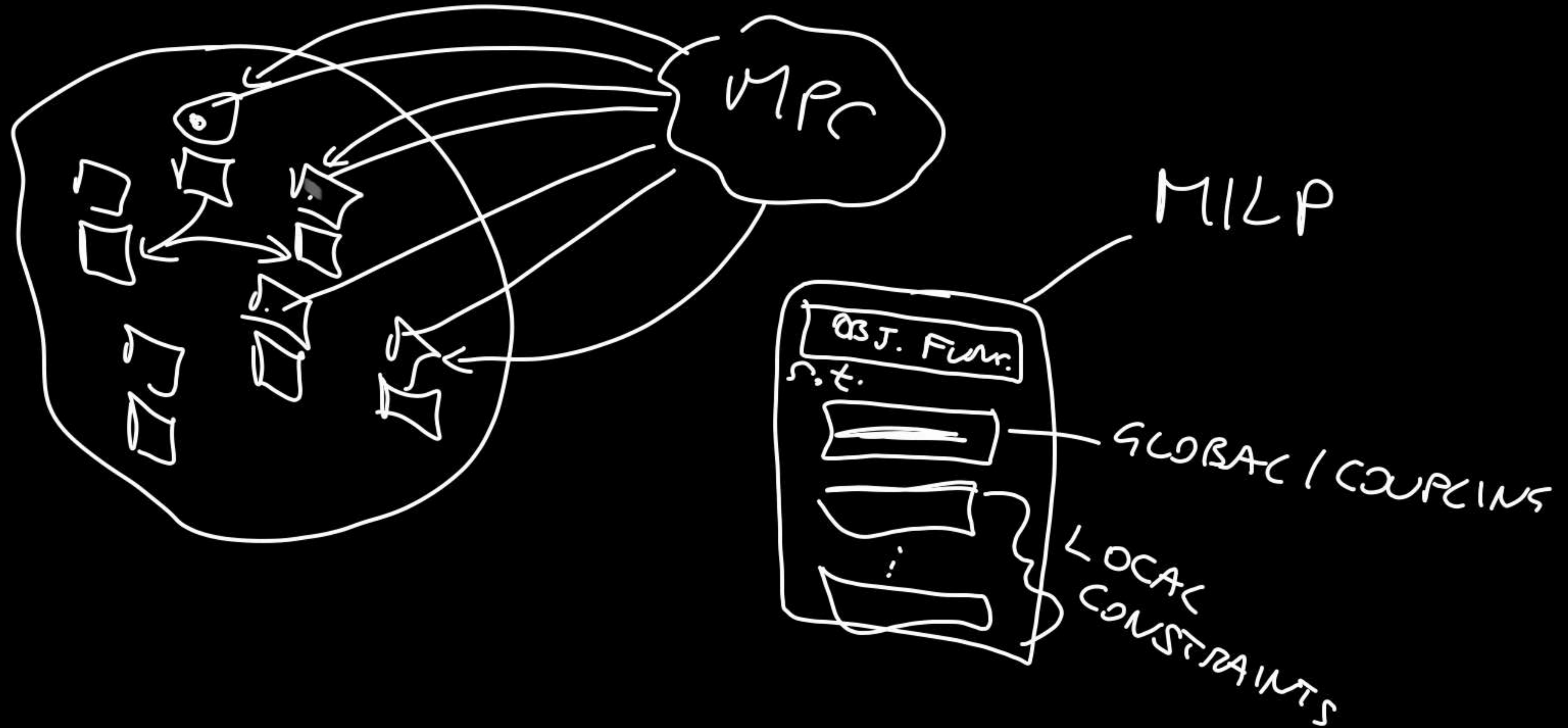
CENTRALISED VS DECENTRALIZED/DISTRIBUTED MPC (3)

- CENTRALIZED: MPC PROBLEM SOLVED IN ONE CENTRAL LOCATION.



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DECENTRALIZED / DISTRIBUTED



ADVANTAGES AND DISADVANTAGES OF MPC:

⑤

PROS :

- ① OPTIMIZATION-BASED APPROACH: PROBLEM GOALS CAN BE DIRECTLY MAPPED INTO THE MPC OBJECTIVE FUNCTION.
- ② CONSTRAINED-CONTROL TECHNIQUE.
- ③ SIMPLE TO EXPLAIN TO INDUSTRY.

CONS:

⑥

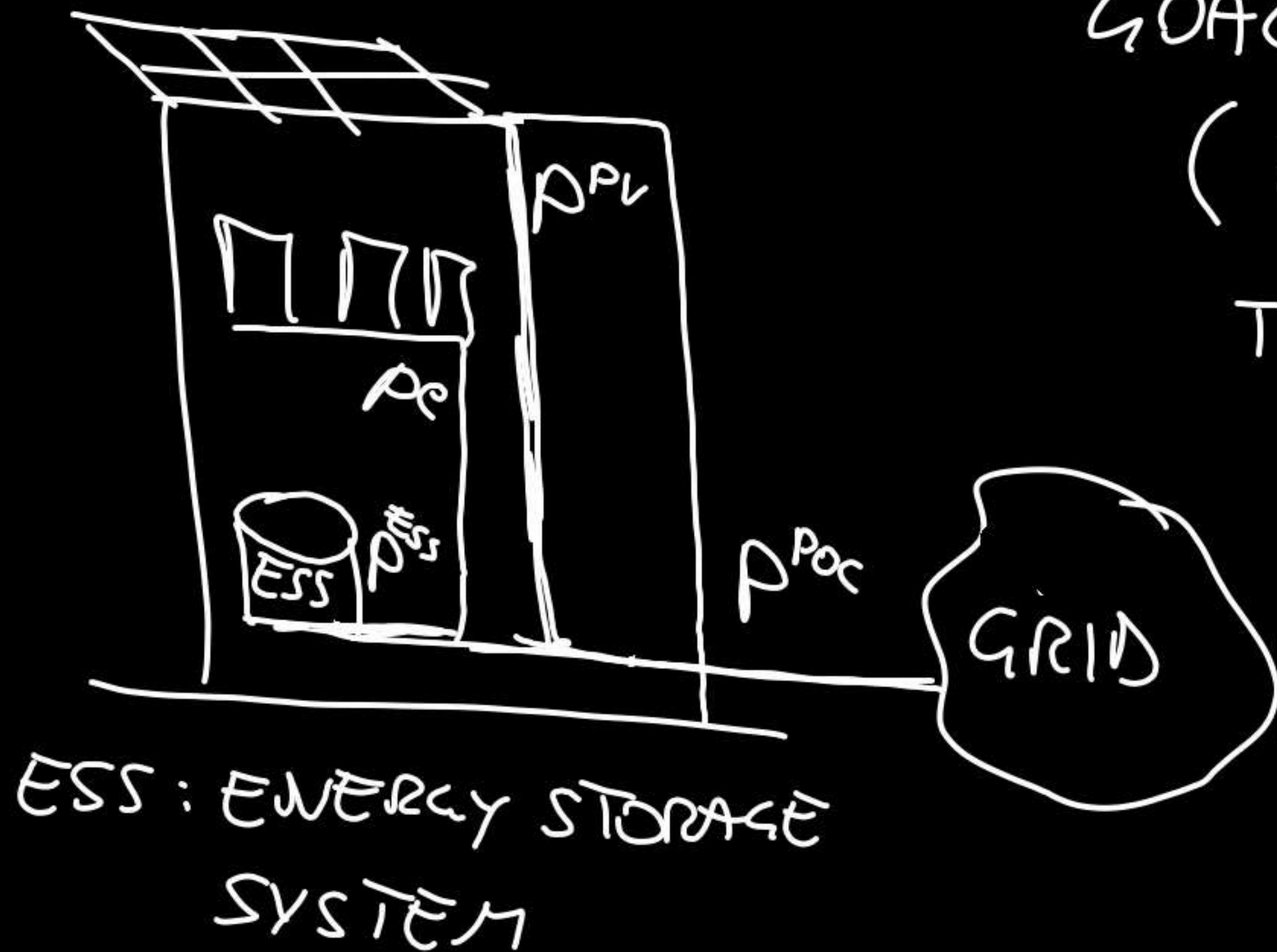
- ① CAN BE COMPUTATIONALLY EXPENSIVE (TOO EXPENSIVE). SOLVING TIME $\ll T$.
- ② YOU NEED A MODEL OF THE SYSTEM.

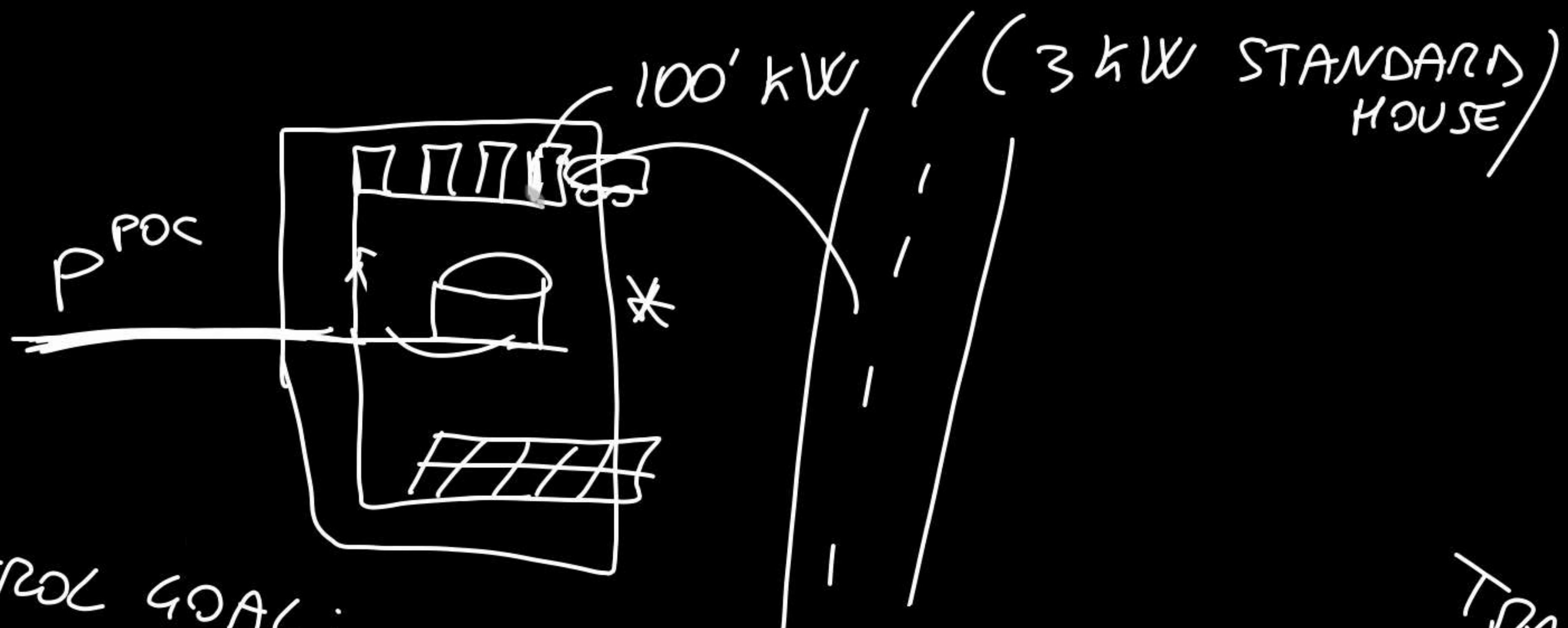
→ SMART GRID/CITIES DOMAINS (ECONOMIC MPC) ⑦

OPTIMIZATION OF ENERGY FLOWS IN A BUILDING:

GOAL: CONTROL THE BUILDING
(ESS, CONTROLLABLE LOADS)
TO:

- MINIMIZE COSTS,
 - USER COMFORT.
- TRADE-OFF





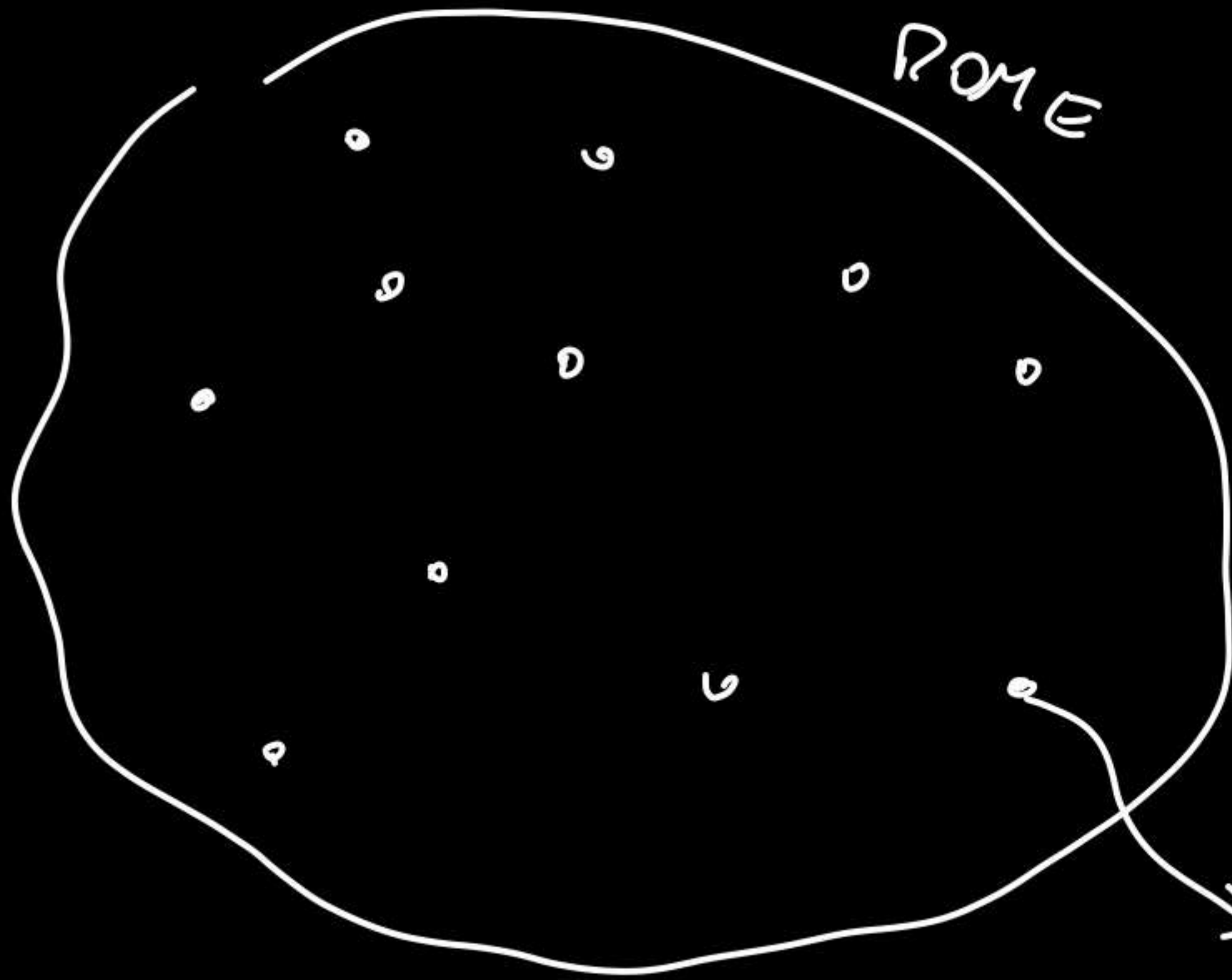
CONTROL GOAL:

- MINIMIZE $P^{POC} \rightarrow$ REGULATE P^{POC}
(SYSTEM GOAL - OPERATOR GOAL)

- USER GOALS: RECHARGE AS FAST AS POSSIBLE.

TRADE-OFF

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GOAL:

- COORDINATE CHARGING TO AVOID GRID ISSUES.

- SATISFY USER PREF,

CAR 100%
BY 8:00

DISTRIBUTED
MPC.

SCHEDULING OF ASSEMBLY LINES:

(10)

ASSEMBLE A PLANE/ROCKET

1000' OF INDIVIDUAL TASKS.

GOALS:

- MINIMIZE TOTAL TIME TO BUILD PLANE, (MAKESPAN)
- RESOURCE BALANCING.

HARD VS SOFT CONSTRAINTS

(11)

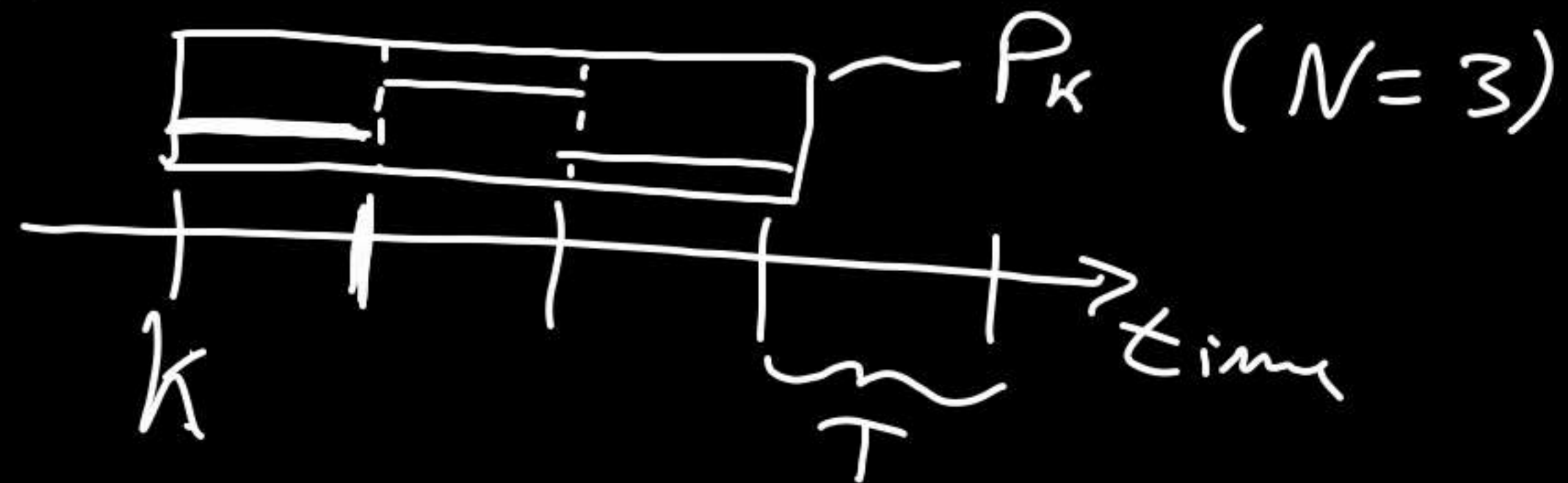
- HARD CONSTRAINTS: IF VIOLATED $\rightarrow$ "PHYSICAL" CONSEQUENCES
THEY ARE INCLUDED AMONG
THE CONSTRAINTS OF MPC PROBLEM

- SOFT CONSTRAINTS: TO ENFORCE "NICE TO HAVE" BEHAVIOUR.

$$- \varepsilon + 2\tau \leq \tau_k \leq 2\varepsilon + \varepsilon$$

$\min \varepsilon$

HOW TO WRITE MPC PROBLEM:

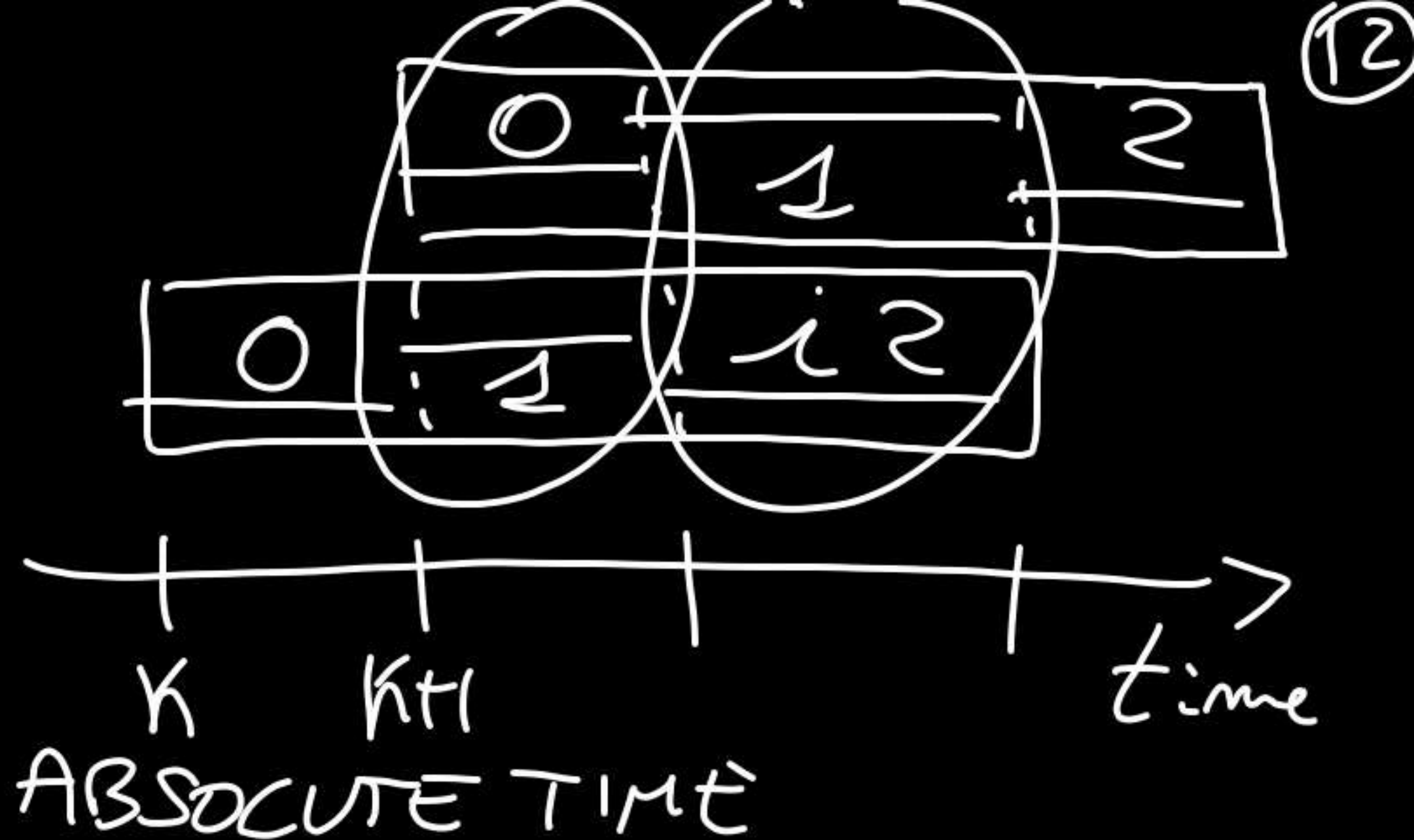


AT EVERY TIME k :

① MEASURE STATE AND TAKE ANY USEFUL INFO

② BUILD AND SOLVE P_k

③ APPLY THE CONTROL COMPUTED FOR THE NEXT TIME INTERVAL AND DISCARD THE OTHERS.



TWO TIME INDICES NOMENCLATURE

(13)

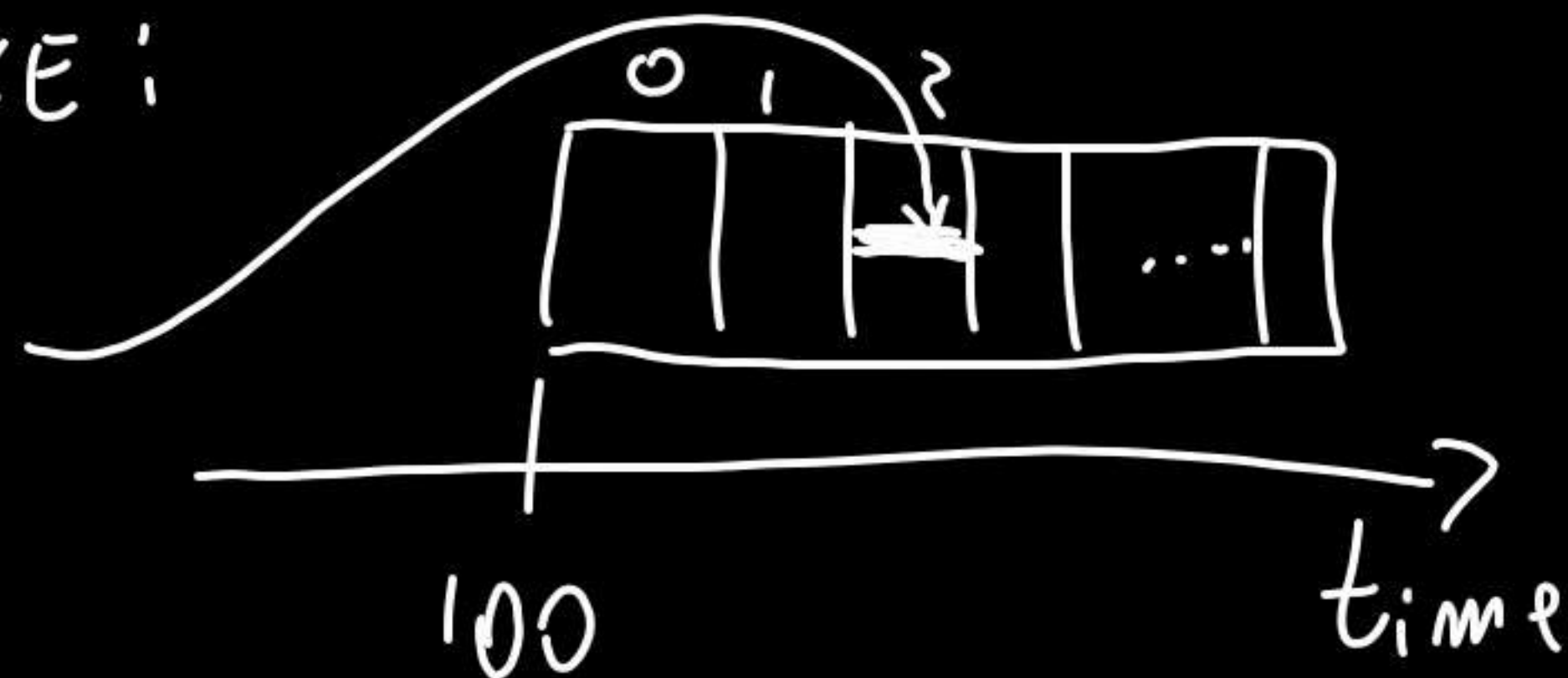
k (ABSOLUTE TIME) TO REFER TO THE TIME WHEN THE MPC PROBLEM IS SOLVED

i (RELATIVE TIME IN THE PREDICTION WINDOW): TO DENOTE TO WHICH INTERVAL OF THE PREDICTION WINDOW THE VARIABLE IN QUESTION REFERS.

x_{ik}, u_{ik}

FOR EXAMPLE:

$u_{2/100}$



$$u_{ik}, x_{ik}$$

$$i = 0, 1, \dots, N$$

WHAT IS u_{0k} ?

(14)

WHAT IS x_{0k} ? STATE OF SYSTEM

AT THE BEGINNING OF FIRST INTERVAL.

$$(x_k, u_{0k}) \xrightarrow{\text{MODEL}} x_{1k}$$

$$x_{2k}$$

$$x_{Nk}$$

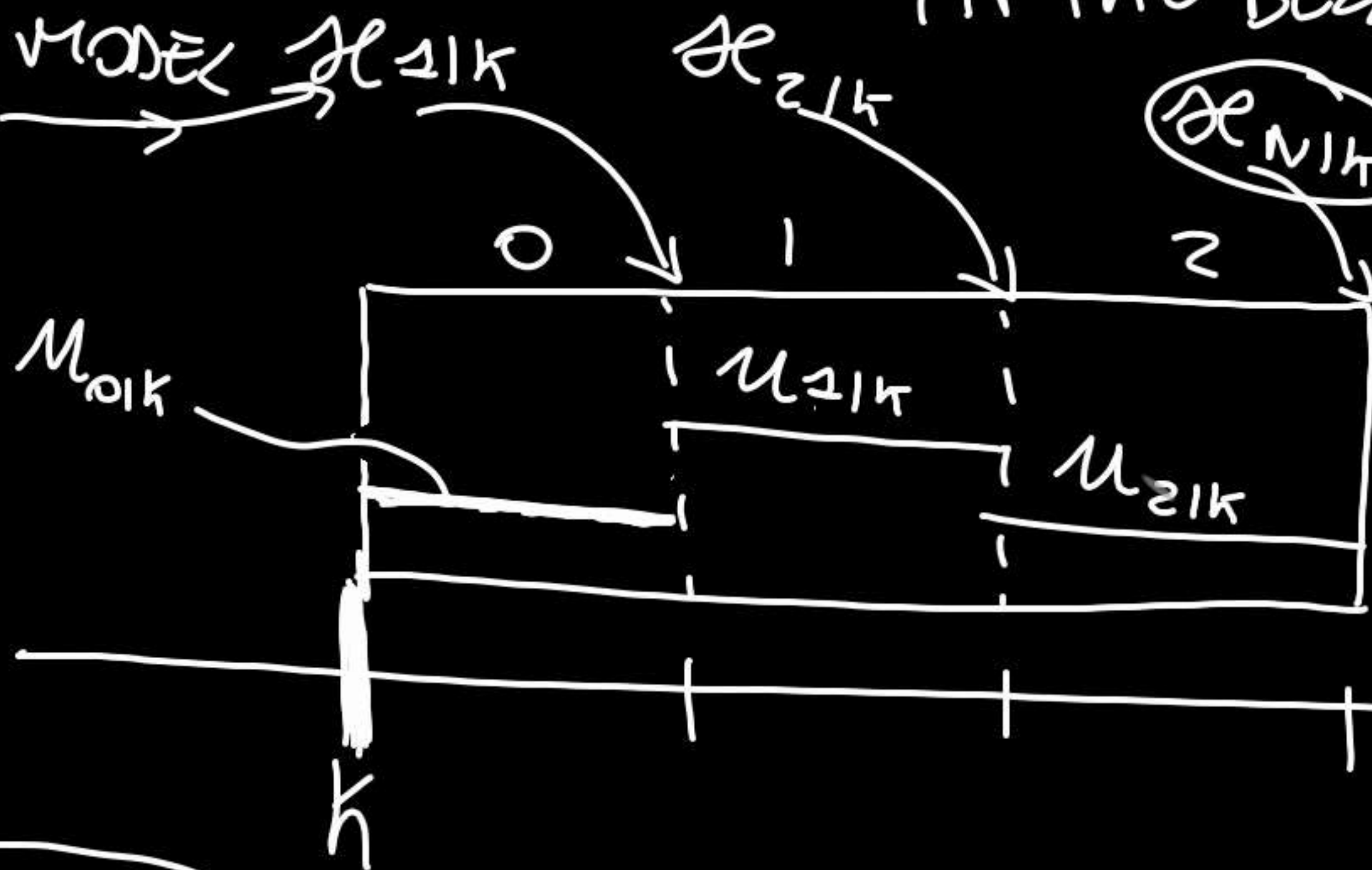
WHAT IS x_{1k} ?

THE PREDICTED STATE

OF SYSTEM AT

BEGINNING OF 2ND

TIME INTERVAL



$$x_k \equiv x_{0k}$$

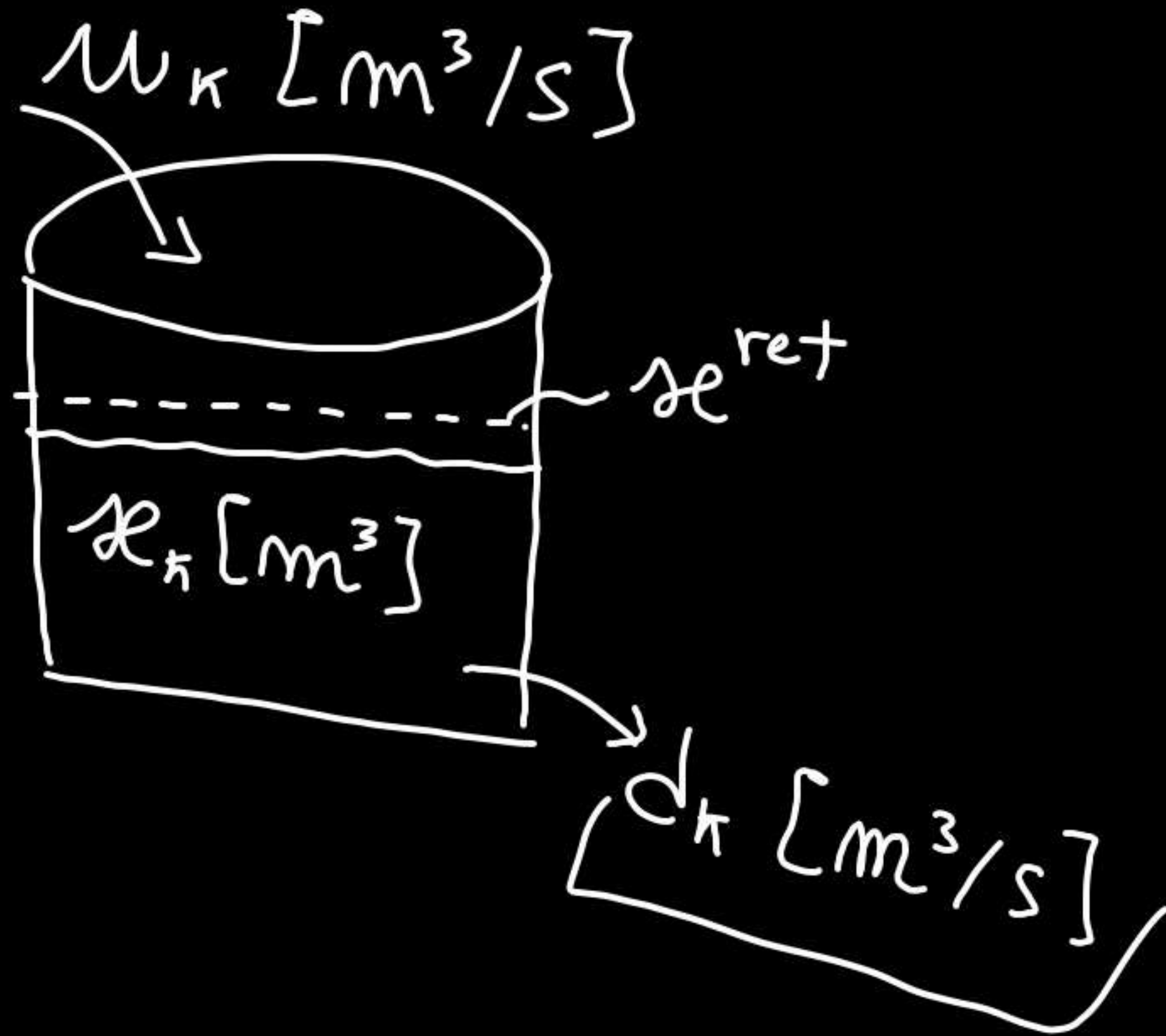
$$x_k$$

IS MEASURED

x_{Nk} TERMINAL STATE

TOY EXAMPLE / FIRST MPC PROBLEM

(15)



d_k : FLOW OF WATER TAKEN AT k .

u_k : CONTROLLED FLOW AT k .

x_k : CURRENT VOLUME OF WATER AT k .

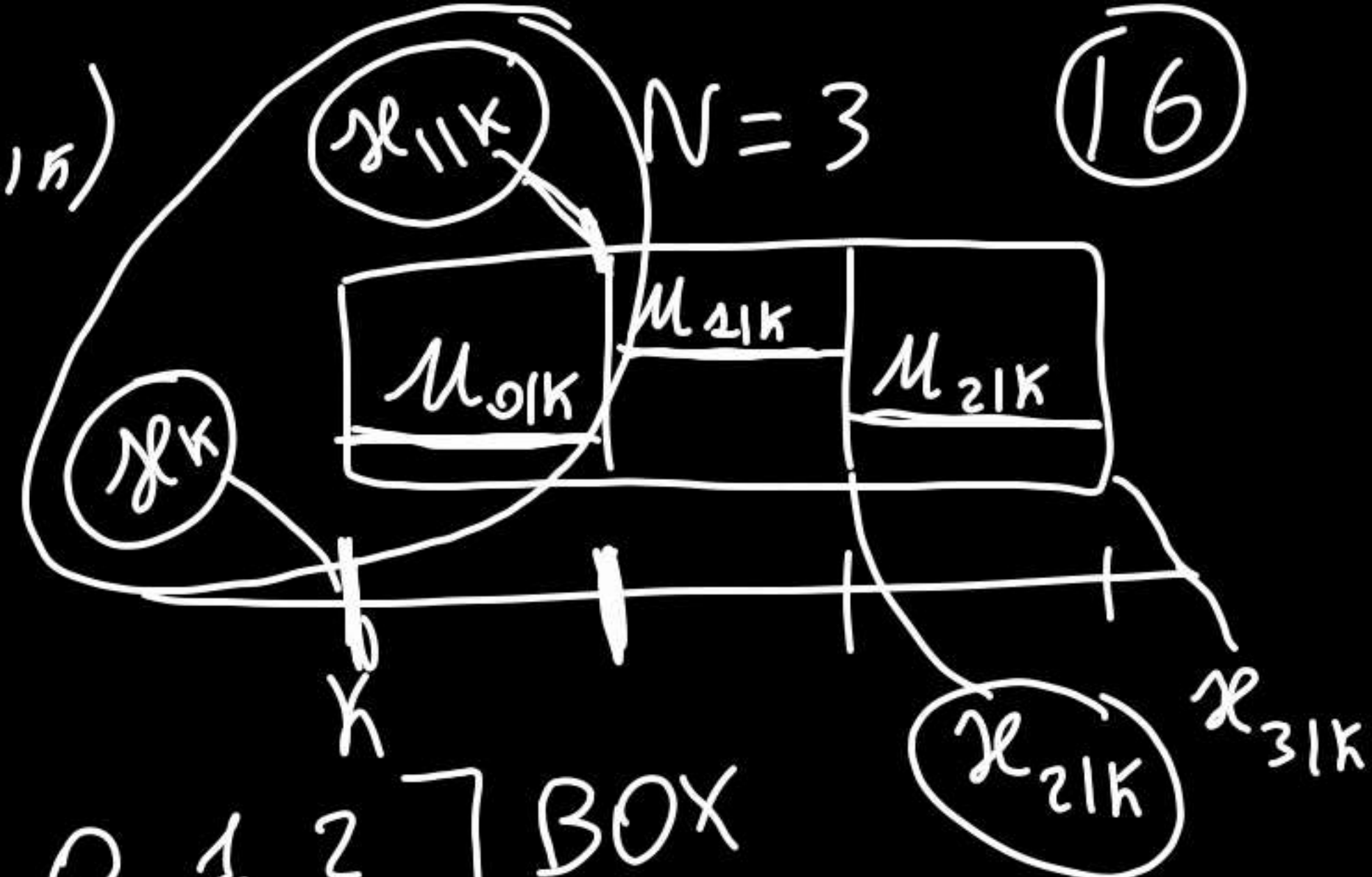
GOAL: KEEP VOLUME x_k CLOSE TO THE REFERENCE.

STATE REGULATION.

MPC FORMULATION ($x_{i|k}, u_{i|k}$) (16)

CONSTRAINTS:

- LIMITATIONS ON x AND u .



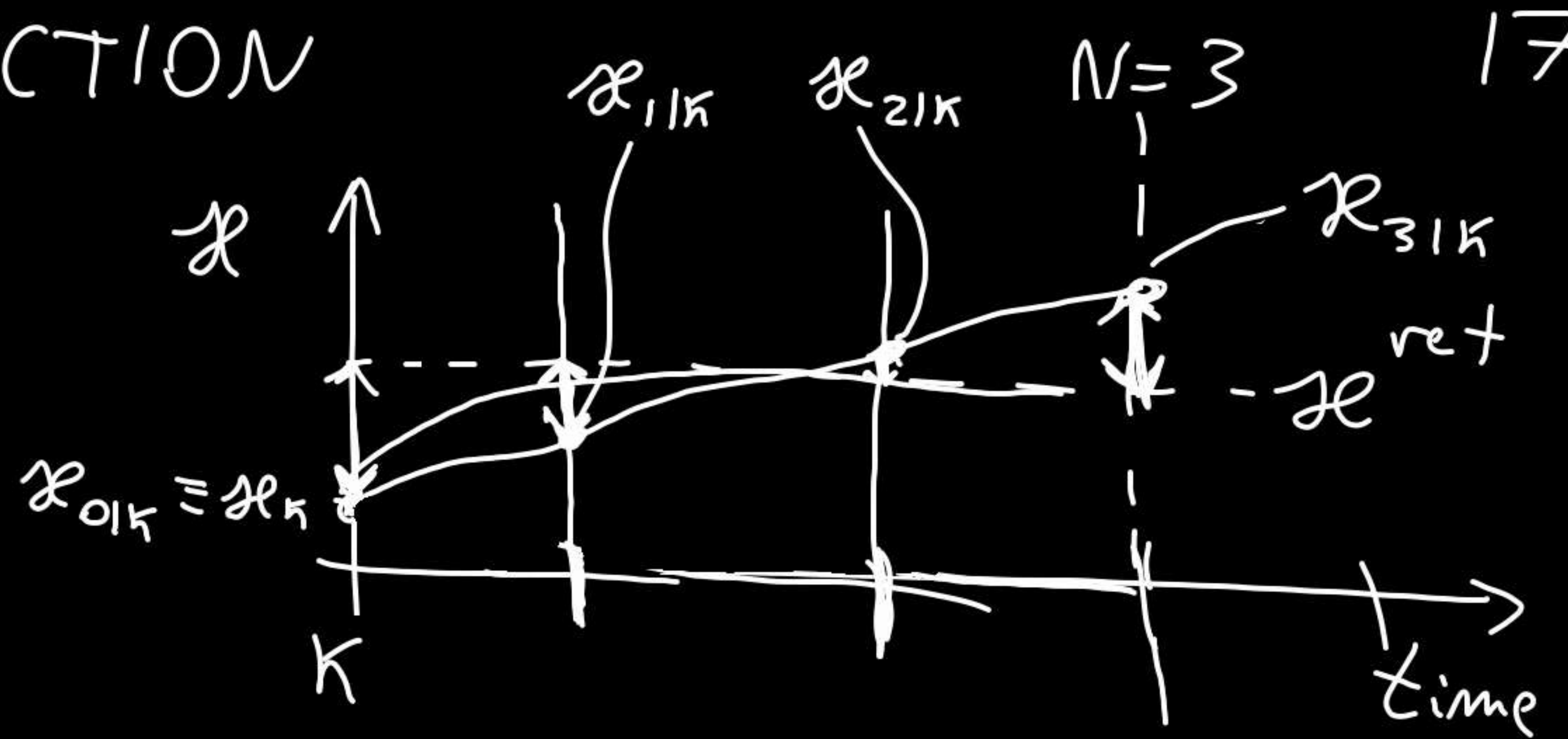
$$\left[\begin{array}{l} 0 \leq u_{i|k} \leq u^{\max}, \quad i = 0, 1, 2 \\ 0 \leq x_{i|k} \leq x^{\max}, \quad i = 0, 1, 2, 3 \end{array} \right] \text{ BOX CONSTRAINTS}$$

$$\rightarrow \left(x_{i+1|k} = \frac{1}{A} x_{i|k} + u_{i|k} \cdot \frac{T}{B} - d_{i|k} \cdot T, \quad i = 0, 1, 2 \right) \text{ DYNAMICS OF SYSTEM}$$

$x_{0|k} = x_k$ (MEASUREMENT OF CURRENT STATE)

TARGET FUNCTION

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$$\min_{u_{0|k}, u_{1|k}, u_{2|k}} \left[\sum_{i=0}^N \underbrace{\left[q \cdot (x_{i|k} - x^{ref})^2 \right]}_{\text{STATE COST}} + \underbrace{\left[b \cdot (u_{i|k})^2 \right]}_{\text{MINIMIZE CONTROL EFFORT}} \right]$$

DYNAMICAL EQUATIONS

CONTINUOUS TIME

DISCRETIZATION

$$\frac{\overset{T}{\sim}}{t \equiv k} \quad \dot{x}(t) \approx \frac{x_{k+1} - x_k}{T}$$

$$\begin{cases} \dot{x}(t) = f(x(t), u(t)) \\ x(t_0) = x(t_0) \end{cases}$$

STATE
EQ.

$$y(t) = g(x(t), u(t)) \quad \text{OUTP. EQ.}$$

NONLINEAR

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ x(t_0) = x_0 \end{cases}$$

$$y(t) = Cx(t) + Du(t)$$

LINEAR SYST.

$$\begin{cases} x_{k+1} = f(x_k, u_k) \\ x_{k_0} \text{ GIVEN} \end{cases}$$

NONLINEAR
SYSTEM

$$y_k = g(x_k, u_k)$$

$$\begin{cases} x_{k+1} = Ax_k + Bu_k \\ x_{k_0} \text{ KNOWN} \end{cases}$$

$$y_k = Cx_k + Du_k$$

